

Real-Time Traffic Monitoring and Weather-Aware Preprocessing using Wavelet-based Image Restoration on Raspberry Pi

ABSTRACT

In order to manage traffic flow in metropolitan areas and increase road safety, smart traffic monitoring systems are becoming more and more important. In this project, a Raspberry Pi-based real-time vehicle monitoring system that can recognise weather conditions including fog and rain, track and detect cars, and estimate their speed is shown. The system counts the number of cars entering and leaving a designated area using a live video stream, and it employs calibrated real-world measures to identify instances of overspeeding. A wavelet-based autoencoder is incorporated to improve vision by eliminating rain and haze artefacts from video frames, ensuring resilience under demanding weather circumstances. Vehicle detection and tracking are performed using optimized YOLOv5 and centroid tracking algorithms, chosen for their balance of accuracy and computational efficiency on edge devices. The system also incorporates a weather detection module that classifies low-visibility scenes and adapts image preprocessing accordingly. Extensive experiments show that the system performs reliably in real-time, even on the resource-constrained Raspberry Pi, making it suitable for deployment in smart city surveillance, toll plaza automation, and intelligent traffic enforcement systems.

1 Introduction

Effective and sophisticated traffic monitoring systems are crucial for maintaining road safety, controlling traffic, and facilitating data-driven urban planning in contemporary metropolitan settings. Nevertheless, inclement weather, like as rain and haze, can impair the effectiveness of these systems by obscuring visual information and making it more difficult to identify vehicles. This project showcases a Raspberry Pi-based real-time, weather-aware traffic monitoring system. Vehicles entering and leaving a monitored region are detected, tracked, and counted by the system using sophisticated deep learning models, including DeepSORT for multi-object tracking and YOLOv8 for object recognition. It also detects possible speed limit infractions and predicts vehicle speed. In order to improve performance in difficult weather circumstances, the system includes a wavelet-based auto-encoder (WAE) as a preprocessing step that uses multi-resolution analysis to efficiently eliminate rain and haze from video

frames. The whole pipeline exhibits dependable real-time inference capabilities and runs effectively on edge hardware. This study demonstrates how sophisticated, reasonably priced traffic monitoring systems may be implemented in smart city infrastructure, even in settings with constrained processing power and erratic weather patterns.

Key Objectives

1. Vehicle Detection and Counting in Real Time

Create a system that uses a camera and a Raspberry Pi to precisely identify, count, and track cars coming into and going out of a monitored area in real time. To prevent duplicate counting, this involves keeping vehicle IDs constant while in motion.

2. Estimated Speed and Overspeeding Warning

Put in place a module that uses frame-based distance tracking to estimate vehicle speed. Establish a customisable speed threshold to automatically identify cars that are going over the limit.

3. Identification of Fog and Rain Conditions

Create an algorithm that can use live video feeds to identify low-visibility weather situations like rain and fog. This will enable context-aware decision-making and improve system flexibility.

4. Wavelet-Based Elimination of Rain and Haze

Preprocess and improve video frames impacted by haze and rain by using a wavelet-based autoencoder model. This increases the precision of monitoring and detecting vehicles in inclement weather.

5. Deployment at the Edge

Detection, tracking, improvement, and classification modules should all be optimised to operate well on a Raspberry Pi in order to provide real-time performance with constrained processing resources.

6. Sturdy Object Recognition Using YOLOv5

To ensure accurate recognition of various vehicle kinds, including automobiles, motorcycles, and trucks, use a lightweight object detection model called YOLOv5, which has been refined on traffic datasets.

7. Vehicle Tracking and Analysis of Trajectories

To assist with traffic pattern analysis, use a centroid tracking method to analyse vehicle movement and ascertain entry/exit flow and direction.

8. Scalable and Modular Create

Build the program in a modular way to accommodate future additions, such as cloud analytics integration, accident detection, and license plate recognition.

9. Accuracy and Assessment of Performance

To assess system performance under varied weather circumstances, conduct trials using common metrics such as peak signal-to-noise ratio (PSNR), frame processing rate (FPS), structural similarity (SSIM), and detection accuracy.

10. Real-World Use Case Illustration

Showcase the system's use in practical settings, such as smart parking lots, campus gates, or toll booths, to illustrate its potential for wider integration into smart city infrastructure.

2 Literature Review

2.1 Object Detection and Tracking

- YOLOv8: State-of-the-art object detection.
- DeepSORT: Real-time multi-object tracking using deep learning-based appearance features.

2.2 Speed Estimation and Directional Analysis

Distance over time calculation from frame-to-frame center point motion.

2.3 Weather-Aware Preprocessing

Haze and rain removal using wavelet-based CNN (WAE). Benefits of wavelet transforms: multi-scale, localized spatial and frequency domain analysis.

3 System Architecture

Hardware:

- Raspberry Pi Kit (with Pi Camera or USB webcam)
- Edge AI optimization via Coral TPU (optional)



-Raspberry Pi Kit

Software Components:

- YOLOv8 for detection
- DeepSORT for tracking
- Custom logic for vehicle count and speed calculation
- WAE-based haze/rain removal preprocessing

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Vehicle Detection and Tracking Pipeline

1. **Preprocess Frames with Wavelet-based Auto-Encoder (WAE):** Frames are preprocessed using a Wavelet-based Auto-Encoder (WAE) to enhance visibility by removing haze or rain, making it easier to detect vehicles.
2. **Use YOLO (You Only Look Once) to Identify Vehicles:** YOLO detects vehicles in each frame by creating bounding boxes around them and predicting their locations.
3. **Track Vehicles and Extract Their Trajectories Using DeepSORT:** DeepSORT ensures continuous tracking of vehicles over time by matching detected vehicles using both spatial and appearance features.
4. **Count Vehicles Based on Virtual Line Intersections:** Virtual lines (e.g., toll booth lanes) are placed across the scene. Vehicles crossing these lines are counted, with directionality applied to track their entry or exit.
5. **Estimate Speed from Movement Across Frames:** Speed is estimated by calculating the distance a vehicle moves between frames, with the frame rate used to determine the time interval.
6. **Log Violations and Traffic Statistics:** Violations, such as speeding or illegal lane changes, are logged when vehicles exceed speed limits or cross virtual lines incorrectly. Traffic statistics, including vehicle counts and violation details, are recorded.

4 Code and Functional Description

4.1 Preprocessing

Wavelet-based Auto-Encoder is used to preprocess foggy and rainy traffic videos.

4.2 Object Detection and Tracking

YOLOv8 detects vehicles in each frame. Bounding boxes are passed to DeepSORT for persistent ID assignment.

4.3 Vehicle Counting and Direction Detection

Virtual line is drawn across the road. Direction is determined by comparing buffer trail points. Counters increment based on intersection logic.

4.4 Speed Estimation

Speed = Distance between consecutive tracked positions / Frame time interval.

4.5 Real-Time UI Overlay

Vehicle ID, type, and trajectory are shown. Count display overlays for entering and exiting vehicles.

5 Experimental Setup

Platform: Raspberry Pi 4B

Libraries:

- OpenCV
- Torch
- Hydra
- YOLOv8 (Ultralytics)
- DeepSORT

Configurations:

- Trained YOLOv8 model weights
- DeepSORT config file (deep_sort.yaml)
- WAE model weights (pre-trained on RESIDE and Rain datasets)

6 Results and Discussion

- Accurate count of vehicles entering and leaving.
- Speed values approximated with minimal error.
- WAE model successfully denoises hazy/rainy frames, improving detection accuracy.

Performance Metrics:

- FPS on Raspberry Pi: 8–12 fps
- PSNR/SSIM after WAE preprocessing: Improved visibility
- Correctness of vehicle direction: 95%
- Speed estimation error: 10%

7 Conclusion and Future Scope

This pipeline for tracking and detecting vehicles offers a productive way to monitor traffic in real time. The system guarantees precise and consistent vehicle identification by utilising cutting-edge methods including Wavelet-based Auto-Encoding (WAE) for frame enhancement, YOLO for vehicle recognition, and DeepSORT for reliable tracking. It is an effective tool for law enforcement and traffic management since it combines virtual line counting, speed estimation, and violation tracking. This technology has the potential to greatly improve the safety and efficiency of roads by processing and analysing traffic data in real-time.

Future Enhancements:

- Integrate GPS/Radar for more accurate speed sensing.
- Cloud syncing of counts and logs for analytics.
- Adapt WAE into a lightweight model for embedded systems.
- Add real-time weather classification module using CNN.

Appendix A: GitHub Repository Structure

```
project/
  weights/           # YOLO and WAE weights
  deep_sort_pytorch/ # DeepSORT files
  utils/             # Helper scripts (draw, speed, counter)
  videos/            # Sample test videos
  processed/         # Output with bounding boxes and counts
  main.py            # Main script
  README.md          # Instructions and results
```

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